

CTF Title: The Leaked Event Photo

Scenario

A photographer was hired to cover a corporate event called Hive Cybersecurity Summit 2026. After the event, one private photo of a VIP guest was leaked online before the official release. The photographer claims:

"I only took the pictures. I did not leak anything."

Investigators recovered one image file from the event media group:

VIP_Backstage.jpg

Your job is to analyze the metadata of this single photo and determine what happened; the camera used, the exact capture time, the original filename, the editing software used, whether GPS data existed, whether metadata was stripped, whether a file was exported, whether timestamps are inconsistent, whether a file came from social media, whether an image was copied to USB, and whether a person's story matches the technical evidence.

Investigative focus areas:

1. EXIF camera metadata
2. XMP editing metadata
3. IPTC copyright and creator fields
4. GPS coordinates
5. File system timestamps
6. Embedded thumbnails
7. Original filename
8. Camera serial number
9. Lens serial number
10. Software and creator tool
11. Compression and image dimensions
12. Hash values
13. ZIP/archive metadata
14. Cloud sync metadata
15. USB connection traces
16. Social media stripping effects
17. Timeline consistency
18. Metadata manipulation signs

Metadata Analysis CTF Questions

Evidence File: `VIP_Backstage.jpg`

Question 1: Camera Identification

What camera model was used to capture the image?

Question 2: Photographer Attribution

Who is the owner or creator of the image according to the metadata?

Question 3: Original Filename

What was the original camera filename before the image was exported or renamed?

Question 4: Decode the Base64 Flag

The `Image Description` field contains an encoded value. Decode it and submit the flag.

Question 5: Capture Time

What date and time was the photo originally taken?

Question 6: Decode the Hex Flag

The `User Comment` field contains a hexadecimal string. Decode it and submit the flag.

Question 7: Editing Software

Which software was used to edit or export the image?

Question 8: Decode the ROT13 Flag

The `Copyright Notice` field contains a ROT13-encoded flag. Decode it and submit the flag.

Question 9: Export Time

What time was the image modified or exported?

Question 10: Decode the URL-Encoded Flag

The `XP Comment` field contains a URL-encoded flag. Decode it and submit the flag.

Question 11: GPS Location

What city and country does the GPS/location metadata point to?

Question 12: Decode the Binary Flag

The `XP Keywords` field contains binary text. Decode it and submit the flag.

Question 13: Metadata Timeline

Compare the original capture time and the modify/export time. What does this tell you about the image?

Question 14: Final Reversed Flag

The `Document Name` field contains a reversed flag. Reverse it and submit the final flag.

Question 15: Final Investigation Conclusion

Based on the metadata, was the image leaked directly from the camera, or was it edited/exported before leaking? Explain your answer.

